Stabilizing Threshold-Based Video Anomaly Detection via Reinforcement Learning

1st Jayapriya D
Department of CSE
PES University
Bangalore, India
jayapriyad@pes.edu

2nd Siya Moghe
Department of CSE
PES University
Bangalore, India
siya.moghe17@gmail.com

3rd Dennis Philip
Department of AIML
PES University
Bangalore, India
dennisphilip.gg@gmail.com

4th Alisha Prakash
Department of CSE
PES University
Bangalore, India
alishaprakash2468@gmail.com

5th Aditya Hubli
Department of CSE
PES University
Bangalore, India
aditya.m.hubli@gmail.com

***Abstract*—Unsupervised video anomaly detection commonly relies on reconstruction-based models trained on normal data, where abnormal events are identified through elevated reconstruction errors. In practice, these error signals are often noisy and sensitive to threshold selection, leading to unreliable anomaly localization. In this work, we study video anomaly detection from a decision-centric perspective and formulate the problem as sequential decision-making rather than static score thresholding. An LSTM-based temporal prediction model is first trained to learn regular motion dynamics and produce frame-level anomaly scores. Reinforcement learning is then used as an adaptive decision layer to dynamically adjust detection sensitivity through threshold control and temporal confidence modulation. Experiments conducted on the Avenue dataset demonstrate that reinforcement learning improves detection stability and robustness without modifying the underlying anomaly scoring model. We further evaluate the anomaly scoring pipeline on the UCSD Ped2 benchmark using deep spatiotemporal features, highlighting the impact of representation strength on frame-level anomaly detection performance.**

***Index Terms*—LSTM, reinforcement learning, anomaly detection, video anomaly detection**

I. Introduction

The rapid growth of video surveillance infrastructure in public and private spaces has created a strong demand for automated systems capable of detecting anomalous events in real time [1], [5]. Anomalies in surveillance videos are typically rare, subtle, and context-dependent, making manual monitoring inefficient and unreliable over long durations [1], [7]. As a result, learning-based approaches that can automatically model normal behaviour and detect deviations have become a central research focus.

II. Related Work

A. Classical Anomaly Detection and Benchmark Datasets

Early work in video anomaly detection focused on modelling normal motion patterns using handcrafted features and statistical learning. One such example is the work by Lu et al. [1], which introduced a sparse reconstruction based framework capable of operating at real time speeds. Evaluated on the Avenue [1] dataset, this approach demonstrated that abnormal events can be detected as deviations from learned normal behaviour. While effective, such methods depend heavily on manually designed features and lack flexibility when faced with complex or evolving motion patterns.

B. Temporal Modelling with LSTM and Autoencoder Architectures

To reduce dependence on hand-crafted features, more recent work has shifted toward sequence prediction models that learn temporal patterns directly from data. Hasan et al. [2] trained an LSTM autoencoder using only normal video samples, where reconstruction error serves as the anomaly score. Building on this, spatiotemporal autoencoders were proposed to jointly capture appearance and motion related information [3] The importance of long-term temporal context for anomaly detection was further highlighted by Luo et al. [4] using LSTM-based models.

C. Deep Spatiotemporal and Object-Centric Approaches

Other than the recurrent architectures, some deep convolutional networks have also been considered for the task of anomaly detection. Sabokrou et al. [5] have described a fully convolutional network for fast detection in a crowded scenario. However, in the case where the anomalies follow objects, such systems prove less helpful.

To mitigate these issues, object-centric solutions emerged for tracking purposes with models independent for each tracked object. Object-centric autoencoders and anomaly synthesis methods [6] illustrate the efficacy of analyzing temporal patterns for individual objects for localization and better explainability in anomaly detection for Avenue-like image sets with abnormalities pointed out by irregular movement of objects.

D. Reinforcement Learning for Anomaly Detection

Reinforcement learning has also been proposed as a new paradigm for anomaly detection owing to the capability to dynamically learn decision changes over time. A systematic review by Arshad et al. in [7] informs about the capabilities of deep reinforcement learning in dealing with sequence and non-stationary anomaly detection tasks. However, most studies based on reinforcement learning aim for direct classification or even ignore sequence tasks altogether in the case of anomaly detection.

III. Dataset and Preprocessing

To evaluate the proposed decision-centric anomaly detection framework under different conditions, we consider two benchmark datasets with distinct characteristics. The Avenue [1] dataset is used as the primary experimental setting to study reinforcement learning–based decision stabilization under noisy anomaly scores. In addition, the UCSD Ped2 [12] dataset is employed as a supplementary benchmark to analyze anomaly scoring behavior when stronger feature representations are used. Due to these differing objectives, separate preprocessing pipelines are adopted for each dataset.

A. Dataset I: Avenue [1]

The Avenue dataset [1] consists of 16 training sequences containing only normal behavior and 21 test sequences that include anomalous events such as running, loitering, and unusual object interactions. Object-level information is extracted using a detection-and-tracking pipeline. Each frame is processed using the YOLOv8 [8] object detector to identify pedestrians and relevant objects.

Detected objects are associated across frames using DeepSORT [9], which combines motion modeling with appearance-based association to maintain stable identities in mostly static-camera scenes. Compared to alternative trackers, DeepSORT produced smoother trajectories with fewer identity switches in our setting.

For each tracked object, motion and appearance features are extracted at every time step. Bounding box coordinates are normalized with respect to frame dimensions, and temporal differences are used to capture short-term motion dynamics. Compact gradient- and colour-based descriptors are also obtained. These features are concatenated to form an object-level representation and aggregated at the frame level to produce anomaly scores.

B. Dataset II: UCSD Ped2 [12]

UCSD Ped2 consists of pedestrian walkway scenes recorded by a static camera. The training set contains only normal behavior, while the test sequences include anomalies such as bicycles and vehicles.

No object detection or tracking is employed for this dataset. Instead, raw video frames are processed directly to extract global spatiotemporal features. Dense optical flow is computed between consecutive frames to capture motion information, and appearance features are extracted using a pretrained ResNet-18 network. These cues are combined to form a deep feature representation for each frame and passed to the same LSTM-based temporal prediction model used in the Avenue experiments.

Pixel-level ground truth masks are converted into frame-level labels by marking a frame as anomalous if any pixel is labelled abnormal.

C. Summary

In summary, Avenue is used to evaluate reinforcement learning–based decision stabilization under noisy anomaly scores using object-centric features, while UCSD Ped2 serves as a complementary benchmark to study anomaly scoring behaviour with stronger deep representations. This separation allows the effects of feature representation and decision adaptation to be analysed independently.

IV. Methodology

Anomaly scoring and anomaly decision-making are treated as two separate components. An LSTM-based temporal prediction model is first used to generate frame-level anomaly scores. These scores are then processed by adaptive decision controllers designed to improve temporal stability and reduce fragmented detections. We look into two methods of decision adaptation. The first being a lightweight feedback driven threshold for frame level models and the second being a RL-based decision controller for object based models. Both methods operate on different levels and thus rely on different LSTMs.

A. LSTM-Based Anomaly Scoring Models

1) LSTM-Ped2: For the UCSD Ped2 dataset, anomaly scoring is performed using a frame-level temporal prediction model. This is a single layer LSTM with hidden size 256, followed by a linear layer for reconstructing frame-level features. The input to the LSTM consists of frame-level spatiotemporal features extracted from RGB appearance and flow representations. The LSTM's training is self-supervised, using normal training frames. During inference, the mean squared prediction error between the predicted and observed features is used as the frame-level anomaly score. Higher prediction errors indicate deviations from learned normal dynamics.

2) LSTM-Avenue: For the Avenue dataset, anomaly scoring is performed using an object-centric temporal prediction model. Objects are detected and tracked across frames. A 44-dimensional feature vector is extracted for each object at every time step. This representation combines motion-related features (e.g., normalized position, velocity, acceleration, trajectory curvature) and appearance descriptors. This is a multi-layer (3) layer LSTM with hidden size 256 and dropout of 0.3. Weights and residual-like context improve temporal modelling. The prediction error produced by the LSTM is aggregated at the frame level by averaging errors across all active tracks, resulting in a frame-level anomaly score.

B. Decision Adaptation Mechanisms

1) Feedback-Driven Adaptive Thresholding (Ped2): For the UCSD Ped2 dataset, a lightweight feedback-driven adaptive thresholding mechanism is used to stabilize frame-level anomaly decisions The anomaly scores are first temporally smoothed using a sliding window average. Then a scalar threshold that is updated based on the current score using a simple reward rule:

$$\text{threshold}_{t+1} = \text{threshold}_t + \alpha \cdot (\text{score}_t - \text{threshold}_t)$$

where α controls the adaptation rate. This encourages the threshold to track the score distribution. It promotes separation between normal and anomalous frames. Detections are triggered whenever the smoothed score exceeds the adaptive threshold. A frame is flagged as anomalous whenever the smoothed score is higher than the adaptive threshold. While simple, this feedback-driven mechanism improves decision stability.

2) RL (Avenue): For the Avenue dataset, a reinforcement learning setup is used to adjust how confident the LSTM is about anomalies. Importantly, the reinforcement learning component does not replace the anomaly scoring model. It operates as a decision-level controller that adjusts detection sensitivity. First, a temporal confidence signal is computed. This is done by comparing short-term and long-term moving averages of the anomaly scores (sudden changes stand out). The action space is intentionally low-dimensional, and the agent learns to balance sensitivity and temporal consistency rather than perform direct anomaly classification. The reward function encourages the agent to amplify sustained anomalous patterns while suppressing transient noise, thereby increasing temporal coherence. On this signal an adaptive threshold is calculated. The RL agent looks as recent confidence values and thresholds to decide a scaling factor. This adjusts the scores, an anomaly is is flagged when the scaled score is above the threshold.

Overall, methodology demonstrates that treating video anomaly detection as a sequential decision problem, rather than a static classification task.

TABLE I
COMPARISON OF LSTM ARCHITECTURES AND DECISION MECHANISMS ACROSS DATASETS

Dataset	**LSTM Type**	**Input Features**	**Decision Mechanism**	**RL Type**
Ped2	Frame-level	Frame reconstruction	Adaptive threshold	RL-inspired
Avenue	Object-centric	Motion + Appearance trajectories	Action selection via DQN	True RL

V. RESULTS

This section shows the performance of the two pipelines.

A. Performance

1) UCSD Ped2: For the Ped2 dataset, the RL inspired thresholding strategy is applied on the frame level LSTM reconstruction scores.

Table II summarizes the quantitative performance.

TABLE II
ANOMALY DETECTION RESULTS ON UCSD PED2 USING ADAPTIVE THRESHOLDING

Metric	Value
AUC (raw)	0.6381
AUC (smoothed)	0.6493
F1-score (RL)	0.643
Precision (RL)	0.916
Recall (RL)	0.495

These metrics show that the RL-inspired adaptive thresholding achieves an F1-score of 0.643, with very high precision (0.916) but lower recall (0.495). Which means that even though the method has few false positives, it misses some anomalous frames.

2) Avenue: For the Avenue dataset, the RL-gated temporal confidence mechanism is applied to the LSTM embeddings. The RL agent scales the temporal confidence signal then applies an adaptive threshold.

TABLE III
FRAME-LEVEL ANOMALY DETECTION PERFORMANCE ON THE AVENUE DATASET.

Method	Precision	Recall	F1-score	ROC–AUC
Adaptive threshold	0.205	0.123	0.153	0.470
RL (Avenue)	**0.241**	0.146	0.182	**0.493**

B. Visual Analysis

This shows the quantitative and qualitative evaluations of the proposed pipelines. Performance metrics are reported first, followed by a visual analysis of anomaly score behaviour.

1) UCSD Ped2: The ROC curve in Fig. 1, confirms the mod7erate discriminative ability. It has an AUC of 0.649 after smoothing.

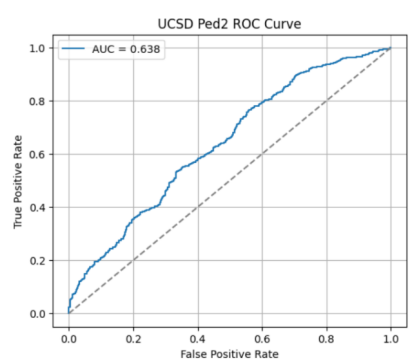

Fig. 1. ROC curve of RL-inspired thresholding on UCSD Ped2.

Further, inspection of Fig. 2, shows that the RL-inspired threshold can track variations in the reconstruction scores. But, short anomalies are often missed (lower recall).

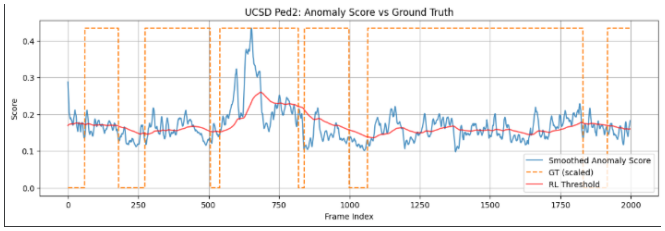

Fig. 2. Anomaly scores versus ground truth on UCSD Ped2.

2) Avenue: From Fig. 3, in deterministic adaptive thresholding, the threshold closely follows short-term fluctuations in the anomaly score distribution. As a result, isolated score spikes frequently trigger false alarms, while brief drops in score magnitude may suppress true anomalies. This behaviour leads to fragmented detections and unstable decision patterns, particularly in sequences containing gradual or prolonged anomalous events.

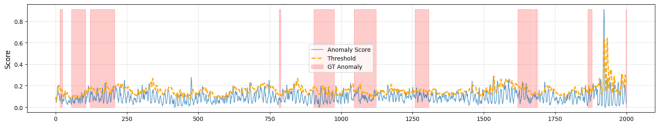

Fig. 3. LSTM-A baseline detection.

From Fig. 4, the RL (Avenue) formulation improves temporal consistency by modulating a confidence signal derived from short and long-term score statistics. Visual inspection shows that transient score spikes caused by normal motion are suppressed, while sustained deviations are amplified. This leads to fewer isolated false positives and more coherent anomaly segments, corresponding to the improved precision and ROC–AUC reported in the results.

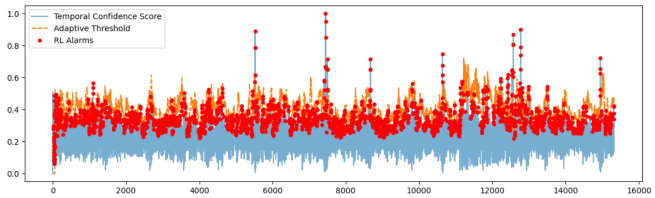

Fig. 4. RL (Avenue) confidence modulation.

Overall, the visual results confirm that reinforcement learning does not alter the underlying anomaly scores, but instead reshapes the decision dynamics over time. The learned policies act as stabilizing controllers, reducing sensitivity to noise and enforcing temporal coherence in anomaly decisions.

VI. Conclusion

This work viewed anomaly detection from a decision-centric perspective. It shifted the focus from static to adaptive thresholding. Experiments on the Avenue dataset shows that reinforcement learning–based confidence modulation leads to improved temporal consistency, higher precision, and better ROC–AUC compared to deterministic adaptive thresholding. On the UCSD Ped2 dataset, the RL-inspired adaptive threshold demonstrated that even lightweight feedback-driven updates can improve score stability when stronger spatiotemporal features are used, but recall remains limited.

Acknowledgment

The authors would like to thank the Center of Cloud Computing & Big Data (CCBD) and Centre for Data Science and Machine Learning (CDSAML), and PES University, Bangalore, India, for their support.

References

[1] Y. Lu, Y. Zhang, and S. Liu, "Abnormal event detection at 150 FPS in MATLAB," in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), Sydney, Australia, 2013, pp. 2720–2727.

[2] M. Hasan, J. Choi, J. Neumann, A. K. Roy-Chowdhury, and L. S. Davis, "Learning temporal regularity in video sequences," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), Las Vegas, NV, USA, 2016, pp. 733–742.

[3] W. Luo, W. Liu, and S. Gao, "Abnormal event detection in videos using spatiotemporal autoencoder," in Proc. Int. Symp. Neural Netw. (ISNN), Chongqing, China, 2015, pp. 3–10.

[4] W. Luo, W. Liu, and S. Gao, "Remembering history with LSTM for anomaly detection," in Proc. IEEE Int. Conf. Multimedia Expo (ICME), Hong Kong, 2017, pp. 439–444.

[5] M. Sabokrou, M. Fayyaz, M. Fathy, Z. Moayed, and R. Klette, "Deep-anomaly: Fully convolutional neural network for fast anomaly detection in crowded scenes," Comput. Vis. Image Underst., vol. 172, pp. 88–97, Jul. 2018.

[6] R. Ionescu, F. S. Khan, M. Georgescu, and L. Shao, "Object-centric auto-encoders and dummy anomalies for abnormal event detection in video," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), Long Beach, CA, USA, 2019, pp. 7842–7851.

[7] K. Arshad, R. A. Ali, A. Muneer, I. Aziz, S. Naseer, N. Khan, and S. M. Taib, "Deep reinforcement learning for anomaly detection: A systematic review," IEEE Access, vol. 10, pp. 1–23, 2022, doi: 10.1109/ACCESS.2022.3224023.

[8] G. Jocher, A. Chaurasia, and J. Qiu, "YOLOv8: Ultralytics next-generation object detection," Ultralytics, 2023. [Online]. Available: Ultralytics YOLOv8 documentation.

[9] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in Proc. IEEE Int. Conf. Image Process. (ICIP), Phoenix, AZ, USA, 2017, pp. 3645–3649.

[10] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "ByteTrack: Multi-object tracking by associating every detection box," in Proc. Eur. Conf. Comput. Vis. (ECCV), Tel Aviv, Israel, 2022, pp. 1–21.

[11] N. Aharon, R. Orfaig, and B. Z. Y. Shapira, "BoT-SORT: Robust associations multi-pedestrian tracking," arXiv preprint arXiv:2206.14651, 2022.

[12] V. Mahadevan, W. Li, V. Bhalodia, and N. Vasconcelos, "Anomaly detection in crowded scenes," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), San Francisco, CA, USA, Jun. 2010, pp. 1975–1981.